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Protein Purification by Affinity Ultrafiltration

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I. INTRODUCTION

Tremendous success in molecular biology and gene engineering has made it possible to express practically any given amino acid sequence and to synthesize practically any desired protein. That success has overshadowed the fact that to be used commercially the expressed protein should be purified to a sufficient degree depending on its final application. The higher the required purity of the protein is the more expensive purification procedure is, mounting to 80% of the overall production costs [1]. The efficiency and economic liability of the purification process determine often the fate of the protein product on the market. Purification of a protein from a fermentation broth generally involves a combination of techniques resolving the proteins according to their size, charge, hydrophobicity, or ability to bind some particular substances (affinity ligands, chelated metal ions, HS-containing substances). These techniques have traditionally been optimized individually rather than as a part of the integrated, continuous process. Moreover, a lot of the development and refining of the purification protocol has been carried out on a laboratory scale with little consideration given to the economy of scaling up.

In order to cut down on costs and improve product recovery in the process consisting of several purification stages, it is critical to design purification techniques that could combine high-resolution power, high-throughput capacity, and process impure feed streams.

Thus, the development of the successful techniques should address two main points. The first imperative requirement is the selection of the protein of

interest from the rest of the proteins and its preferential, ideally complete, accumulation into one phase, while the rest of proteins accumulate in the other phase. The second step is the mechanical separation of these two phases arranged in a way allowing high flow rates and the ability to process turbid solutions with minimal mechanical entrapment of impurities in the phase enriched with target protein. Let us address each of these requirements in turn.

II. SELECTIVE PARTITIONING OF THE TARGET PROTEIN—USE OF AFFINITY LIGANDS

Many proteins possess intrinsic ability to bind some particular substances. For example, many enzymes have strong affinity toward their substrates or substrate-similar molecules; protein inhibitors that selectively bind proteins to which they are targeted and antibody-antigen pairs present nearly perfect selectivity of binding. All these types of interactions combined under the name of affinity interactions are successfully used for selective binding of the target protein from the crude extract [2]. Whenever the natural binding force of the protein is insufficient for the purification process, the target protein could be modified genetically via fusing different affinity tails or tags in order to facilitate the purification process [3]. Thus, affinity ligands coupled to a matrix present an efficient tool capable of recognizing the target protein and binding it, resulting in the enrichment of the matrix compared to the aqueous phase containing crude extract.

III. SEPARATION OF TWO PHASES, ENRICHED WITH AND DEPLETED OF TARGET PROTEIN

The solution of the second problem, separation of enriched phase from the depleted crude extract, was traditionally realized in the format of the column chromatography. The ligand-bearing matrix is represented by chromatographic adsorbent and packed into a column with a porous bottom. The starting mixture is pumped through the column, providing enough contact time for binding of the target protein by adsorbent to take place. The column format suffers a few drawbacks. High flow rates through the column result in significant flow resistance of the column and require high pressures to maintain desired flow rates. Moreover, the solution applied on a column should be completely free of particulate material. Even minute “haze” frequently appearing in concentrated protein solutions could be deadly for the performance of the column, which works as a depth filter accumulating particulate material between the beads of the adsorbent. The final result is a complete blocking of the flow through the column. Expanded bed chromatography has emerged as an attempt to use particulate-containing feeds

in chromatography [4]. On the other hand, new techniques were developed where the separation of enriched and depleted phases was carried out in other than column format. In affinity precipitation the target protein is accumulated in the polymer precipitate; the latter is separated by centrifugation or filtration [5]. In affinity partitioning in aqueous polymer two-phase systems, target protein is enriched in one of the liquid phases; the other one is separated using separating funnel or other techniques for liquid-liquid separation [6].

Ultrafiltration separates molecules on the basis of their size and shape. For maximum separation efficiency, there should be 10-fold difference in the sizes of the species to be separated [7]. Thus, to make ultrafiltration applicable for protein purification, the target protein should be somehow included in a complex of at least 10 times larger than the impurities, which have usually the same size range as target protein. This could be achieved by binding the target protein to a macroligand, a soluble polymer, or a microparticle-bearing affinity ligand specific toward the target protein. On the other hand, the macroligand should not be too big and precipitate on gravity. It should remain in a soluble or suspended state enabling easy mass transfer and avoiding any aggregation. The high-molecular-weight soluble polymers or microparticles (latexes) seems to be objects of choice for affinity ultrafiltration.

IV. ULTRAFILTRATION

Membrane filtration in general and ultrafiltration in particular is an established technology in biochemical and pharmaceutical industry. Ultrafiltration membranes have pore sizes of 0.001 to 0.1 μm . Because of the low or negligible osmotic pressure of macromolecules, ultrafiltration operates at relatively low pressures, typically 2 to 5.5 bar. Ultrafiltration is an efficient process for removing macromolecules, colloids, emulsified oil, endotoxins, pyrogens, viruses, and bacteria. It is widely used in the food industry, for treating industrial wastewater, and processing of drinking water. In 1996, the market for ultrafiltration membranes and systems was nearly \$530 million [7]. Developments in polymer chemistry and module construction have resulted in the availability of an array of ultrafiltration membranes of high chemical and physical stability combined with high fluxes, for different scales of operation. Hydrophobic membranes can be modified to increase flux and reduce fouling. The modification techniques includes the following: reacting of the base polymer with hydrophilic pendant groups before casting; entrapping a hydrophilic moiety before casting; blending polymer before casting; modifying membrane surface or charge; grafting the ceramic membrane surface [7].

Filtration is most effective when operated as a tangential flow (also termed crossflow filtration) such that the flow of the feed is parallel to the membrane

surface and is recirculated rapidly enough to prevent concentration polarization.

V. AFFINITY ULTRAFILTRATION

A. Concept

In order to combine the high-volume processing capability of membrane filtration and at the same time increase its specificity, the concept of affinity ultrafiltration was developed. This technique is based on a principle that target protein and impurities, when free in solution, pass through the ultrafiltration membrane, whereas when the target protein is bound to a macroligand, the resulting complex is retained by the membrane. The impurities are washed out after the binding of the target protein to the macroligand. When elution buffer is introduced, the target protein dissociates from the macroligand and is transported in a purified form through the membrane with the flux (Figure 1) [8–14].

This process can be operated as a batch, semibatch, or continuous process. The respective sizes of the target protein and macroligand alongside with the

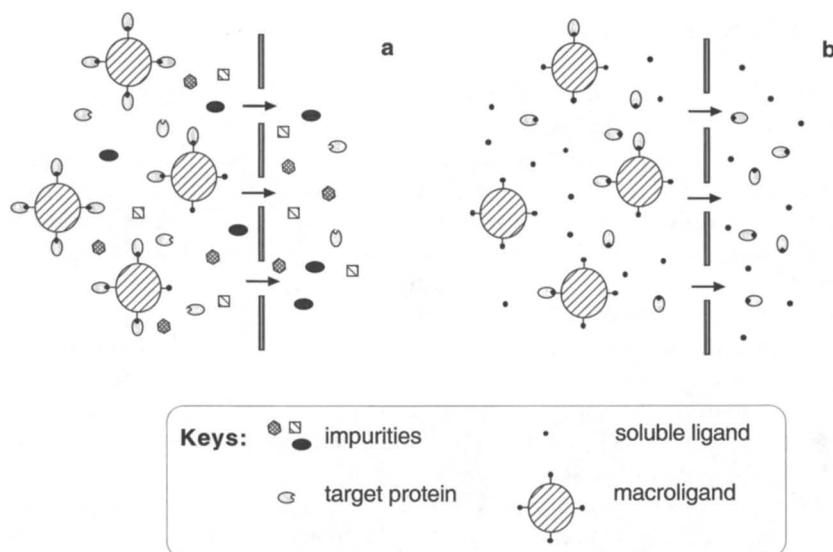


Figure 1 Principle of affinity ultrafiltration: (a) binding of the target protein and washing out impurities; (b) dissociation of the complex target protein-macroligand mediated by free ligand and transport of the purified target protein through the membrane.

cutoff of the membrane determine the efficiency of the purification procedure. The membrane should prevent the leakage of macroligand and, at the same time, should be highly permeable to the free protein and the impurities. Therefore, the study of the individual behavior of the macroligand and target protein prior to the affinity step is highly recommended [15].

The choice of the membrane is dictated not only by its cutoff properties but also by the chemical nature of the membrane material. Macroligand could adsorb or deposit as a layer on its surface resulting in reduced filtration rates. Membrane could adsorb target protein as was demonstrated during isolation of urokinase by affinity ultrafiltration using copolymer of N-acryloyl-*m*-aminoben-zamidine and acrylamide as a macroligand [8]. Urokinase, with a molecular weight of 31,000, was not detected in permeates through polysulfone Millipore membranes with cutoff 10,000 and 100,000. However, the enzyme passed with 95% recovery through the 100,000-cutoff Amicon YM membrane made of cellulose acetate, suggesting that urokinase was bound to polysulfone membrane.

Protein-protein interactions in the crude could also contribute to the complicated ultrafiltration patterns. During urokinase purification from human urine, high-molecular-weight material present in the crude formed a complex with the enzyme, preventing its passage through 100,000-cutoff membrane. The addition of 2 M NaCl resulted in dissociation of the complex and 70% recovery of urokinase in the permeate [8].

B. Binding

As all protein purification techniques based on selective binding of the target protein, affinity ultrafiltration protocol is divided into four steps: binding, washing, elution, and regeneration of the adsorbent, in this case, macroligand.

Binding of the target protein to the macroligand could be realized either (1) by mixing crude extract with macroligand in the chamber at the same side of the membrane, or (2) by pumping crude extract and macroligand solution/suspension at the opposite sides of the membrane, letting the target protein to diffuse through the membrane to the macroligand-containing chamber where the binding takes place.

The first alternative is more straightforward and constructively simple, but it requires preliminary clarification of the feed. Most of the studies reported so far on affinity ultrafiltration were done in this mode. The second mode is a more general approach and could be directly integrated with, for example, a fermentation broth or complex crude extracts. Here the membrane forms a protective barrier for the ligand against particulate material present in the crude extract [9]. α -Amylase from *Bacillus licheniformis* was purified with 95% recovery by the latter approach using crosslinked starch slurry as the affinity adsorbent [16].

The ligand content in the macroligand affects the affinity ultrafiltration procedure. The binding capacity initially increases with the increase in ligand content of the macroligand, but reaches a plateau at a high ligand content as ligands become sterically inaccessible for protein binding. A similar phenomenon has been observed during affinity chromatography [9].

C. Washing

Washing of the complex target protein-macroligand is usually performed in the diafiltration mode such that the volume of the retentate is maintained constant by adding the washing buffer at a rate equal to the filtration rate [9].

To avoid leakage of the target protein from the complex with macroligand, affinity ultrafiltration requires higher binding strength of the target protein to the macroligand than affinity chromatography. The protein molecule dissociated from the chromatographic matrix during washing has high chances to bind again due to the high volumetric (in the chromatographic column) concentration of the ligands. The protein molecule encounters many interactions on its way through the column, binds to the matrix once again, and remains inside the column. The total volumetric concentration of the ligands in affinity ultrafiltration is lower than in affinity chromatography, and the protein molecule dissociated from the macroligand has much lower chances to meet other ligands when drawn by the flux through the membrane. Thus, it is desirable to use high-affinity ligands for affinity ultrafiltration. According to Ling and Mattiasson [10], if the binding between the protein and macroligand is assumed to be in equilibrium, the binding constant should be $>10^6 \text{ M}^{-1}$. However, for affinity crossflow filtration of human serum albumin using Cibacron Blue agarose as a macroligand, Herak and Merrill [17] showed that release of the target protein during the washing step has very slow kinetics and essentially proceeds under nonequilibrium conditions. This observation opens an opportunity for using affinity ligands with somewhat lower binding strength than 10^6 M^{-1} . One more factor could be also favorable during washing of bound multisubunit proteins, namely, multisite attachment. High local concentration of the ligands in macroligand facilitates multisite binding of these proteins, especially when soluble macroligands are used. Flexible polymer backbone of the macroligand can adapt in solution conformation that allows simultaneous binding of a few ligands to a multisubunit protein. The effect of polymer flexibility on multisite polymer attachment to the protein was demonstrated recently in affinity precipitation using Cu-loaded copolymers of N-isopropylacrylamide and 1-vinylimidazole [18].

D. Elution

The elution of bound protein is achieved by changing the buffer so that the binding interaction is unfavorable and the target protein is released from the complex

with macroligand. As in other affinity techniques, elution is usually achieved either by use of a competitive affinity ligand or by altering pH or ionic strength. The first alternative is attractive, since it is performed under conditions that preserve the biological activity of the target protein. The drawback might be the high cost of using competitive affinity ligand. Nonspecific elution at extreme pH or high salt concentrations is cheap but may be harmful to the target protein. Extreme pH values or high salt concentrations could also adversely affect the properties of the ultrafiltration membrane or macroligand itself.

Affinity ultrafiltration is operated in the mode close to complete mixing, and there is no buffer front as in affinity chromatography. Affinity between the protein and the macroligand changes gradually as the elution buffer replaces the washing buffer. The important aspect of this step is to avoid dilution of the released target protein with excessively large volumes of buffer [9]. The precise selection of elution conditions is very much dependent on the system concerned.

The eluate containing the purified product is very dilute after affinity ultrafiltration. The additional ultrafiltration step could be useful resulting in concentration of the product and at the same time removal of competing ligand when the latter was used for protein elution.

E. Regeneration

After the elution step, the free macroligand should be regenerated and reequilibrated in the binding buffer. Regeneration may include washing out proteins, that were nonspecifically bound to the macroligand and were not eluted with the elution buffer. High salt concentrations or extreme pH values used for washing out nonspecifically bound proteins could be detrimental for both macroligand and the membrane. Hence, the macroligand should be developed with regard to minimal nonspecific interactions with the proteins present in the crude extract. The reequilibration of the macroligand is achieved in a separate membrane filtration step where the elution buffer or washing solution is replaced with the fresh binding buffer by diafiltration.

VI. CHOICE OF MACROLIGAND

The choice of macroligand is critical for the applicability of affinity ultrafiltration. The general requirements to the macroligand are the following:

1. Selective binding of target protein with minimal nonspecific interaction with the other proteins present in the crude extract
2. High capacity, i.e., binding large amounts of target protein per gram of the macroligand
3. Chemical stability, i.e., ability to withstand many cycles without leak-

- age of the ligand and chemical destruction of a soluble polymer or microparticle
4. Mechanical stability, of microparticles, i.e., ability to maintain their size and shape under high shear forces resulted in the liquid flow during pumping
 5. Meeting legal requirements, e.g., being nontoxic, nonpyrogenic, etc.
 6. Cost efficiency

By and large, finding commercially available macroligand is difficult. This limitation is overcome by “tailoring” the macroligand; that is, a ligand molecule is coupled either to a water-soluble polymer or a water-insoluble carrier that does not itself bind proteins.

A. Water-Soluble Macroligands

The advantages of water-soluble macroligands lie in the high yields obtained from ligand coupling, the rapid control of the amount immobilized, and easy complexation (no mass-transfer limitations) with the target protein. High binding capacities per gram of the macroligand could be achieved. Protein binding takes place under equilibrium conditions, resulting in quantitative complexation [9].

Macroligands based on soluble polymers have much smaller size than macroligands based on insoluble carriers. When used in affinity ultrafiltration, soluble macroligands require much smaller pores, resulting in much lower flux rates; accordingly, a longer processing time or a larger membrane surface is needed. To provide sufficient resolution in affinity ultrafiltration, the size of the complex of target protein with the macroligand should differ significantly from the size of protein impurities. Most of the soluble proteins have molecular weight <100,000, so the polymer carrier should be of the molecular weight >100,000, preferably in the range of 1,000,000. Aqueous solutions of polymers of that high molecular weight are often **very viscous** even at relatively low concentration due to the entanglement of macromolecular chains. Both reversible and irreversible fouling can be a significant problem in ultrafiltration of polymeric solutions. On the other hand, soluble polymer is not likely to be degraded by attrition.

The requirement of molecular weight in the range of 10^5 – 10^6 limits the choice of water-soluble polymer carriers to only a few options—namely, polysaccharide-based polymers (dextran, starch) and poly(acrylamide) based polymers (Table 1). The chemistry of attaching ligands to carbohydrate polymers is well established, and high-molecular-weight dextran was used as ligand carrier in a number of studies [19–22]. Unfortunately, separation of trypsin and α -chymotrypsin using trypsin specific conjugate, dextran with coupled soybean trypsin inhibitor, faced significant problems due to the nonspecific adsorption of the proteins to the polymer carrier.

Table 1 Macroligands Used in Affinity Ultrafiltration

Carrier	Size	Ligand	Target protein	References
Soluble macroligand				
Poly(acrylamide)	MW >100,000	<i>m</i> -amino-benzamidine	trypsin	15,55
Poly(acrylamide)	MW >100,000	<i>m</i> -amino-benzamidine	urokinase	8
Dextran	MW 2,000,000	protein A	IgG	19
Dextran	MW 2,000,000	estradiol	Δ_{5-4} 3-ostosteroid isomerase	22
Dextran	MW 2,000,000	<i>p</i> -amino-benzamidine	trypsin	21
Dextran	MW 2,000,000	soybean trypsin inhibitor	trypsin	20
Insoluble macroligand				
Silica nano particles	0.012 μ m	Cibacron Blue	alcohol dehydrogenase, lactate dehydrogenase	10
Liposomes	0.027 μ m	biotin	avidin	29,30
Polystyrene core-shell latex particles	0.25–0.4 μ m	Cibacron Blue	bovine serum albumin	32
Liposomes	0.06 μ m	<i>p</i> -amino-benzamidine	trypsin	31
Cell parts of <i>Streptococcus</i>	1 μ m	surface protein G	bovine IgG	56
<i>Saccharomyces cerevisiae</i> cells	5 μ m	surface carbohydrate groups	concanavalin A	12
Polystyrene core-shell microspheres	2.0–7.7 μ m	endopectin-lyase	polyacid, Eudragit 100–55	35
Starch granules	1–10 μ m	Cibacron Blue	alcohol dehydrogenase	13
Superose beads	11–15 μ m	Cibacron Blue	human serum albumin	57
Superose beads	11–15 μ m	Cibacron Blue	lysozyme	57
Sepharose beads	45–165 μ m	Cibacron Blue	human serum albumin	17,57
Sepharose beads	45–165 μ m	Cibacron Blue	lysozyme	57
Sepharose beads	45–165 μ m	concanavalin A	horseradish peroxidase	57
Agarose beads	45–165 μ m	protein A	IgG	57
Agarose beads	not stated	<i>p</i> -aminobenzyl-1-thio- β -D-galactopyranosid	β -galactosidase	28
Sepharose beads	45–165 μ m	heparin	lactoferrin	56
Sepharose beads	45–165 μ m	protein G	bovine IgG	56

Poly(acrylamide)-based macroligands proved to be more successful for trypsin purification [8,15,23]. The poly(acrylamide)-based macroligands are synthesized by copolymerization of acrylamide with a monomer, N-acryloyl-*m*-aminobenzamidine, which has a benzamidine group specifically binding to the trypsin active site. The mechanism of trypsin inhibition by macroligand seems to be competitive with binding constant dependent on the *m*-aminobenzamidine content in the macroligand. At high *m*-aminobenzamidine content the inhibition capacity of the polymer decreased. The optimal benzamidine content in the macroligand, maintaining a balance between the capacity of the macroligand and strength of binding was around 17 mol%. The macroligand with molecular weight $>10^5$ (fractionated using ultrafiltration through a membrane with 100,000 cutoff) was used for the separation of a model trypsin/ α -chymotrypsin mixture. At optimal conditions, 98% pure trypsin with 90% recovery was obtained in a batch process [24]. Trypsin binding to the polymer proceeds very fast, eliminating the need of preliminary incubation of the crude extract with the macroligand. With the reuse of the same polymer in a continuous process and using benzamidine as the eluent, trypsin was purified from crude porcine pancreatic extract with 97% purity and 77% yield. The decrease in polymer concentration from 0.75% to 0.5% resulted in decrease in trypsin retention from 90% to 54%, suggesting that bound trypsin became more susceptible to the shear forces with the decrease in polymer concentration.

It was shown important to maintain the pH value around 8 during affinity ultrafiltration. Both trypsin and α -chymotrypsin passed slowly through the membrane at low or high pH values. Hence, acidic conditions used for trypsin elution from benzamidine-containing chromatographic matrices could not be used for elution during affinity ultrafiltration. Benzamidine (50 mM) or arginine (1 M) solutions were used instead for trypsin elution [25]. The polymer solution is stable for months and trypsin when bound to macroligand is stabilized against autolysis [26].

The same soluble macroligand was used for urokinase purification from a model mixture containing peroxidase and urokinase and from crude human urine [8]. The bound urokinase could be eluted easily using a buffer solution containing 100 mM benzamidine and 100 mM NaCl. Purification of urokinase from crude human urine was complicated by the fact that when this material was ultrafiltered through a 100,000-cutoff membrane, only 5% of the urokinase was detected in the filtrate. Apparently, there was an aggregation between urokinase and some high-molecular-weight materials to form a complex. When saline solution (2.0 M NaCl) was added to the system at the same rate as the filtrate, about 70% of urokinase was detected in the filtrate. The overall yield of the enzyme in affinity ultrafiltration procedure was 49% and sevenfold purification was achieved.

The recent advances in polymer chemistry resulted in the appearance of new water-soluble polymers. Some of them could be of interest as potential carri-

ers in affinity ultrafiltration. Dendrimers are a new class of polymers, which emerged recently. They have hyperbranched treelike structures of a very rigid nature [27]. Some unique properties of the dendrimers make them attractive as carriers for macroligand production in affinity ultrafiltration. The rigidity of the structure improves the selectivity of the ultrafiltration; no “wormlike” squeezing of the linear polymer through membrane pores could take place. High and easily regulated density of reactive groups at the outer surface of the dendrimers combined with the rigidity allows tuning spatial adjustment of ligands to provide all the advantages of multisite binding of the target protein. Finally, whereas the viscosity of the linear polymer increases logarithmically with the molecular weight, viscosities of dendrimers reach plateaus as the molecular weight increases, making it possible to create high volumetric ligand concentration without macroligand solution being too viscous. With new polymers available, one could expect the revival of the interest to affinity ultrafiltration as well as to other techniques of protein purification based on the target protein binding to a soluble macroligand.

B. Water-Insoluble Macroligands

By definition, insoluble macroligands are much larger than soluble ones, providing higher resolution force between the complex target protein-macroligand and impurities. Particles as large as those used in affinity chromatography could be applicable for ultrafiltration. Agarose and starch particles have been found to be relatively easy to pump through the cross-filtration device with minimal fouling of the membranes [17]. However, particle degradation occurred with time. Particles of a rather rigid highly crosslinked gel, Sepharose CL-6B, were found to be fractured 5% after 24 h of circulation while 30%–40% particles were damaged in just 3 h for a soft gel, Affi-Gel Blue [17]. Particles of a commercially available affinity gel, *p*-aminobenzyl-1-thio- β -D-galactopyranoside (PABTG) agarose, were used for purification of β -galactosidase from *E. coli* homogenate by continuous affinity recycle extraction (CARE) process (28). When dealing with chromatographic materials as ligand carriers, it is no longer necessary to operate with ultrafiltration membranes. The introduction of microfilters gave much better fluxes and thus allowed larger volumes to be processed per unit time.

High volumetric concentration of the ligand is advantageous for the affinity ultrafiltration process, increasing total capacity of the system and concentration of the target product in the eluate. When the ligands are coupled to insoluble carrier, the available surface of the particles sets a limit on the amount of ligand. The larger the particles, the less the surface area they have. Thus to increase surface area one should use either small particles or large porous particles, e.g., agarose-gel particles. In the later case the binding and/or desorption kinetics could be slow due to the diffusional limitations resulting in the obligatory incuba-

tion of the macroligand with a crude extract in a separate vessel for binding to take place or a very dilute eluate.

One of the first studies on affinity ultrafiltration with small insoluble macroligand employed heat-killed cell of *Saccharomyces cerevisiae* (mean diameter, 5 μm) as a macroligand for the purification of concanavalin A (molecular weight 71,000) from jack bean extract. The target protein is specific for carbohydrates and binds to the available carbohydrate ligands exposed on the surface of the yeast cells. The mixture of the cells and crude extract was pumped through a hollow-fiber unit with 1,000,000-cutoff at a flow rate of 500 mL/min. High glucose concentration (150 g/L) was used to dissociate the affinity complex. The glucose was partially removed by passing the permeate from the dissociation step through a membrane with 35,000 cutoff, while the yeast cell suspension was washed from the glucose in the regeneration step. Concanavalin A in a highly purified form was obtained with an overall yield of 70% [12].

Alternatively, small particles (diameter of 12 nm and a surface area of 200 m^2/g) of fumed silica with immobilized Cibacron Blue were used for purification of alcohol dehydrogenase from yeast and lactate dehydrogenase from porcine heart [10]. The ligand density was found to be 34 $\mu\text{mol/g}$, which is equivalent to 0.2 $\mu\text{mol}/\text{m}^2$ and comparable to the density of 0.33 $\mu\text{mol}/\text{m}^2$ for an NAD derivative coupled to silica beads and used in HPLC columns.

Due to their small size, the silica particles formed a stable colloid suspension. The binding kinetics was similar to those obtained for binding in free solution, with binding and dissociation occurring almost instantaneously. Thus the process could be run continuously without the risk of incomplete binding.

Although filtration rate has been greatly improved as compared to previous work, it was still the limiting step in the process. The recovery in the elution step has been shown to follow an exponentially decreasing curve as expected for unbound protein. An alternative to increasing the filtration rate is to decrease the working volume. In this case a higher percentage of particles could be used [13].

Insoluble macroligand should not be necessarily based on a solid carrier. The use of affinity-modified liposomes or vesicles formed by phospholipids for affinity ultrafiltration seems to be an interesting approach [29–31]. Unimolecular liposomes with an average diameter 200 to 700 Å possess an extremely large surface area; e.g., 300-Å-radius liposomes have a specific surface area of 100 m^2/mL of liposomes which in terms of area per carrier volume is approximately the same as for porous silica. The hydrophilic surface of the liposomes minimizes the nonspecific protein adsorption and results in a rather strong repulsion between liposomes. Hence, the membrane polarization and plugging during ultrafiltration are reduced.

Liposomes have historically been used to study the structure and function of biological membranes and as biocompatible drug carriers. The chemistry of liposome modification is well developed. Affinity ligands could be covalently

attached to the liposome surface, either coupling the molecule to suitable functional groups on the outer surface, or incorporating derivatized phospholipids into the membrane during liposome formations. Liposomes are sufficiently stable for this type of modification.

The first approach, a surface modification, was used for purification of trypsin. Liposomes comprised of dimyristoyl phosphatidyl choline, cholesterol, and dimyristoyl phosphatidyl ethanolamine with amino groups at the surface were modified with diglycolic anhydride and *p*-aminobenzamidine was coupled to the terminal carboxy groups via carbodi-imide procedure. The equilibrium binding constant between trypsin and immobilized ligand was shown to be dependent on the ligand density of the liposome surface. The ligand density on the liposome can be carefully controlled by simply varying the amounts of phospholipids during liposome preparation. Bound trypsin was eluted from the liposomes by the trypsin inhibitor benzamidine. Trypsin was purified from a trypsin/chymotrypsin mixture and from porcine pancreatic extract. A recovery yield from the crude mixture of 68% was obtained with a trypsin purity of 98%. Loss of trypsin during washing was thought to be due to the relatively low equilibrium binding constant of $1.7 \times 10^4 \text{ M}^{-1}$ between trypsin and immobilized ligand. The affinity-modified liposomes were stable in the complex mixture and retained their trypsin binding capacity after multiple adsorption/elution cycles over a 1-month period [31].

The second approach, an incorporation of a modified phospholipid into liposome during formation, was used for avidin purification from avidin model mixtures with lysozyme or cytochrome *c*, and also from its naturally occurring source, hen egg white. Biotinylated dimyristoyl phosphatidylethanolamine was synthesized by coupling biotinyl-N-hydroxysuccinimide ester to phosphatidylethanolamine. The liposomes comprised of dimyristoyl phosphatidylcholine and biotinylated dimyristoyl phosphatidylethanolamine. Only $\sim 1\%$ of the total biotin in the system was accessible for avidin binding. The equilibrium binding constant was determined to be $6 \times 10^7 \text{ M}^{-1}$ as compared to 10^{15} M^{-1} for the free biotin-avidin in solution. The drastic decrease in binding efficiency was attributed to the steric hindrance between the avidin molecule and the liposomes as the biotin was immobilized directly on the liposome surface; however, the binding strength was sufficient for purification purposes. Crude hen egg white extract was subjected to ion exchange chromatography and the eluate was processed in affinity crossflow ultrafiltration system resulting in 96% pure avidin with 80% recovery of its initial activity [30].

Polystyrene core-shell latex particles of submicron size were prepared by a seeded emulsion polymerization and composed of a hard polystyrene core surrounded by a hydrophilic shell consisted of crosslinked glycidyl methacrylate, acrylamide, and vinyl sulfonate [32]. Triazine dye, Cibacron Blue, was used as an affinity ligand and coupled directly to the latexes after the hydrolysis of surface epoxide groups. The degree of coupling was controlled by the ionic strength of

the reaction medium. The adsorption of bovine serum albumin to the latex was proportional to the ligand density and reached 70% of the maximal within 1 min. Desorption was accomplished using 2 M NaCl. Complete desorption required up to 5 min. A relatively slow binding/desorption kinetics prevented using continuous crossflow systems. Centrifugal filtration with 0.2- μ m pore diameter membrane was used instead. The functionalized core shell particles show a high colloidal stability and a low nonspecific adsorption.

Water-soluble polymers are often used as stabilizers of colloid systems [33]. The polymer-stabilized latex could be used as affinity latex when a target protein has affinity for the polymer stabilizer. For example, concanavalin A binds efficiently to poly(methyl methacrylate) latex synthesized using dextran as the polymer stabilizer [34]. The functional microspheres (2.0 to 7.7 μ m diameter) for the reversible binding of endopectin-lyase were prepared by free-radical dispersion polymerization of styrene using polyacid Eudragit L 100–55 (1:1 statistical copolymer of methacrylic acid with methylmethacrylate) as the polymer stabilizer [35]. The enzyme binds to the microspheres at pH 4.7 and is eluted quantitatively and fast at pH 6.6. The separation of the microspheres was carried out using fast filtering system with 1.2- μ m pore size. When bound to the microspheres, the enzyme retains about 60% activity, indicating that an active site is not involved directly in the binding process. Binding to the microspheres improves storage stability of the enzyme.

The examples presented undoubtedly show that affinity ultrafiltration could be successfully used for protein purification, provided the proper macroligand is developed, while the selection of ultrafiltration system presents no problem as many of them are available at the market. The recent achievements in the chemistry of microparticles resulted in the development of many macroligands capable of reversible binding/elution of a large variety of proteins. These macroligands have not been yet used in ultrafiltration mode but there are no plausible restrictions, why they could not be used for affinity ultrafiltration of proteins. The macroligands, which are suitable for but have not been used in ultrafiltration mode, are reviewed in the following section.

C. Potential Macroligands for Affinity Ultrafiltration

A straightforward way of macroligand synthesis is a direct covalent coupling of the ligand to appropriate microspheres or latexes. There are right now commercially available latexes with a variety of properties. For example, Bang Laboratories Inc (Fishers, IN) produces microspheres with sizes from 20 nm to 1 mm, carrying hydroxyl, sulfonate, amino, carboxy groups, or preactivated with chloromethyl, epoxy, and aldehyde surface groups. Preactivated latexes require only mixing with a ligand in a buffer of the optimum pH for covalent coupling. Parking area, i.e., area occupied by a single active group, varies from a few to a few

hundred square Å, making possible the choice of the optimal ligand density of the macroligand and/or the surface properties with minimal nonspecific protein adsorption.

On top of the numerous latexes available for the synthesis of macroligands, quite a few tailor-made latexes with different ligands were developed. DNA oligomers were immobilized by coupling to epoxy-groups of the latexes comprised of styrene copolymer with glycidyl methacrylate crosslinked with divinylbenzene. The affinity latex was used for purification of DNA-binding transcription factor ATF/E4TF3 from HeLa cell crude nuclear extracts [36]. The latex particles with bound protein were separated by centrifugation and elution was achieved with 1.0 M KCl. At least eight polypeptides with molecular masses 116,000, 80,000, 65,000, 60,000, 55,000, 47,000, 45,000, and 43,000 were copurified. Affinity-purified polypeptides stimulated transcription *in vitro* from a promoter, in which ATF/E4TF3-binding sites were present [37]. Protein kinase II was later shown to copurify with ATF/E4TF3 factor [38]. The enzyme does not bind to the DNA latex but only to the ATF moiety of the ATF/E4TF3 factor.

Latex particles with a single stranded DNA were produced by coupling double-stranded DNA. More than 80% of the double-stranded DNA was coupled through the single-stranded protruding ends. The particles were then treated with alkali or heated to denature double-stranded DNA. The resulting single-stranded DNA particles allowed selective and efficient isolation of complementary RNA from total cellular RNA [39], or rapid and sensitive detection of specific genome DNA, separating normal and point mutant DNA [40,41].

TATA-binding protein from yeast was immobilized via spacer on the latex of the same structure by converting epoxy groups to amino, coupling ethyleneglycol diglycidylether followed by its hydrolysis, and activating of hydroxy groups formed using tosyl chloride. The affinity latex was used for the purification of human transcription factor from an enriched protein fraction of HeLa cell nuclear extracts [42]. A common transacting factor, Ad4-binding protein, was purified from nuclear extracts of bovine adrenal cortex when polymerized Ad4 sequences were coupled to this latex [43]. A peptide Gly-Arg-Gly-Asp-Ser was coupled to the same latex via ethyleneglycol diglycidylether spacer, and the affinity latex was used for purification of receptor proteins from octylglucoside extracts of human platelets. Western blotting with monoclonal antihuman platelet GP IIIa antibody indicated that the purified proteins were GP IIb/IIIa [44]. Arg-Gly-Asp-Ser-carrying latex was used not only for purification of the respective receptor from cell membrane, but also for cell activation [45].

Carboxylated polystyrene/poly(acrylamide) latexes were used for immobilization of bovine serum albumin (BSA) and purification of anti-BSA antibodies [46]. Centrifugation was used to separate latex particles from the supernatant in all the above-mentioned purification procedures.

Recently, Kawaguchi and coworkers [47] proposed a versatile technique for affinity latex production. Hydrophilic monodisperse latex particles are composed of acrylamide, methacrylic acid, methylenebisacrylamide (crosslinker), and nitrophenyl acrylate. The nitrophenyl acrylate presents a reactive moiety allowing a variety of chemical reactions for ligand coupling as well as direct coupling of biopolymers (protein, peptide, DNA) via amino groups (Figure 2).

The latexes are inherently stable; i.e., the latex suspension does not precipitate on gravity. High centrifugal forces are required to separate latex from a supernatant. The latexes composed of styrene, N-isopropylacrylamide and glycidyl methacrylate, or methacrylic acid showed thermosensitivity originated from the thermosensitive nature of N-isopropylacrylamide; that is, at the temperatures above critical, the latex particles flocculate and could be separated from the supernatant by precipitation on gravity or low-speed centrifugation. BSA was covalently immobilized on these latexes and used for purification of anti-BSA antibodies from antiserum. The increase in temperature to 40°C resulted in immediate flocculation of the latex with bound antibodies and easy separation from the supernatant [48].

Another approach for mechanical separation of latex particles from a supernatant was proposed by Kondo et al. [49]. Affinity latex was prepared by immobilizing human γ -globulin onto carboxylated latex composed of polystyrene and poly(acrylamide). A fusion protein (ZZB1B2) of IgG and albumin-binding domains was expressed in *Escherichia coli* and purified by the affinity latex. In a polyethylene glycol (PEG)/potassium phosphate aqueous two-phase system affinity latex with bound recombinant protein was partitioned into the PEG-rich phase, while cells and cell debris were displaced into the salt-rich bottom phase.

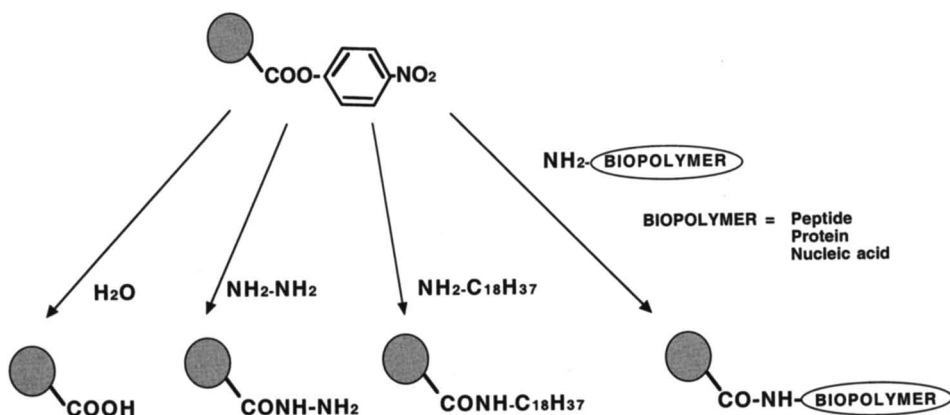


Figure 2 Ligand coupling to nitrophenyl acrylate-containing latex.

The separation of microspheres from supernatant could be achieved using magnetic properties of the microspheres produced of conventional Fe_3O_4 pigment covered with poly(vinyl alcohol). Immobilized *m*-aminophenylboronic acid allows specific binding of glycosylated Hb [50]. High density of the affinity particles formed by poly(vinyl alcohol)-coated perfluorocarbon emulsions allows separation by gravity but requires special construction of adsorber with uprising flow to keep the particles suspended. The continuous purification of malate dehydrogenase from unclarified homogenate of *Saccharomyces cerevisiae* was achieved when a triazine dye, Procion Red HE-7B, was immobilized on the perfluorocarbon particles [51].

The above-mentioned examples do not exhaust all the possible ways of latex synthesis and latex derivatization applicable for the production of affinity macroligands for protein purification. Apart of protein purification, affinity latexes are used widely for analytical purposes, especially in immunoanalysis [52], and less frequently, for immobilization of enzymes [53]. The synthetic procedures and ligand coupling to the latexes used for analytical purposes are reviewed by Kawaguchi [54].

Membranes are available in many materials and with almost any pore size. The choice of membrane is thus not a limitation. What has been limiting the development of affinity ultrafiltration has rather been lack of suitable ligand carriers and laborious and cumbersome coupling procedures. The recent development in polymer chemistry has dramatically changed that picture.

VII. CONCLUSION

Affinity ultrafiltration has not found yet a wide commercial application. It remains mainly the object of academic study. The commercial availability of ultrafiltration equipment of different designs and productivities combined with a variety of macroligands, which are already developed or could be easily synthesized using available carriers, make affinity ultrafiltration an attractive alternative to other protein purification techniques. The possibilities of quickly processing large solution volumes and easy scaleup that make the process automatic, and the feasibility of operating the same unit for purification of different proteins when using different membranes/macroligands constitute the obvious advantages of affinity ultrafiltration and justify hopes for its coming wide application in protein purification.

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